

# Hub

## Modern E-Learning Platform — Full Technical Documentation

Framework: Laravel 12 (PHP 8.3) + PostgreSQL

Frontend: Tailwind CSS v4 + Alpine.js v3

Repository: [github.com/anis3669/learnhub](https://github.com/anis3669/learnhub)

Report Date: April 2025

3

USER ROLES

26

VIEWS

13

DB TABLES

50+

ROUTES

# Executive Summary

## PROJECT OVERVIEW

LearnHub is a comprehensive, production-ready e-learning platform built on Laravel 12 and PostgreSQL. It provides a complete digital learning ecosystem for three distinct user roles — Students, Teachers, and Administrators — each with a dedicated portal, dashboard, and feature set tailored to their responsibilities.

The platform delivers HD video streaming, auto-graded quizzes, a gamification engine (points & badges), a merge sort-powered competitive leaderboard, course-specific discussion forums, and rich analytics dashboards. Every component is built with clean architecture principles, efficient algorithms, and a modern UI design.

## TOTAL USERS SEEDED

13

1 Admin · 2 Teachers · 9 Students · 1 Demo Account



4

16 Lessons · 2 Quizzes · 9 Questions · 36 Options



ENROLLMENTS CREATED

24

Students enrolled across all seeded courses



8

Unique badge types with custom criteria and icons





 **Demo Credentials**

Admin: admin@learnhub.com · Teacher: teacher@learnhub.com · Student: student@learnhub.com — all share the password **password**. Top-scoring student: emma@learnhub.com (920 pts).

## Technology Stack

| Layer             | Technology                         | Version | Purpose                                                                                          |
|-------------------|------------------------------------|---------|--------------------------------------------------------------------------------------------------|
| Backend Framework | Laravel                            | 12.x    | MVC framework, routing, middleware, Eloquent ORM                                                 |
| Runtime           | PHP                                | 8.3     | Server-side scripting language                                                                   |
| Database          | PostgreSQL                         | 15+     | Primary relational database (hosted on Replit)                                                   |
| CSS Framework     | Tailwind CSS                       | 4.x     | Utility-first responsive styling                                                                 |
| JS Framework      | Alpine.js                          | 3.x     | Reactive UI components (sidebar, quiz timer, dropdowns)                                          |
| Build Tool        | Vite                               | 6.x     | Asset bundling and hot module replacement                                                        |
| Auth & RBAC       | Laravel Breeze + Spatie Permission | -       | Authentication scaffolding + role-based access control                                           |
| Templating        | Blade                              | -       | Laravel's compiled template engine                                                               |
| Fonts             | Inter (Bunny CDN)                  | -       | Primary UI typeface                                                                              |
| PDF Generation    | Dompdf                             | 3.x     | Server-side HTML-to-PDF conversion                                                               |
| Version Control   | Git + GitHub                       | -       | Source control — <a href="https://github.com/anis3669/learnhub">github.com/anis3669/learnhub</a> |
| Hosting           | Replit                             | -       | Cloud development and deployment environment                                                     |

### Key Architecture Decisions

#### Monolithic MVC Architecture

Laravel's standard MVC structure keeps concerns separated without the overhead of a separate API layer. Blade templates communicate directly with controllers via Eloquent models.

#### Role-Based Access Control (RBAC) via Spatie

The Spatie Laravel Permission package manages all role definitions and middleware guards, keeping authorization logic out of controllers. Middleware aliases (role:student, role:teacher, role:admin) are registered globally in bootstrap/app.php.

#### Alpine.js for Lightweight Reactivity

Alpine.js handles all in-page reactive UI — sidebar collapse/expand, quiz navigation and countdown timer, flash message dismissal — without the overhead of Vue or React.

#### Vite for Optimized Asset Pipeline

Vite compiles and minifies Tailwind CSS and Alpine.js into versioned bundles (app-[hash].css / app-[hash].js), with Laravel's @vite directive injecting the correct files automatically.

# Database Design

The database consists of 13 tables with clearly defined relationships. PostgreSQL was chosen for its robustness, ACID compliance, and support for complex joins used in the leaderboard and analytics queries.

| TABLE              | KEY COLUMNS                                                                                  | RELATIONSHIPS                                                                                          |
|--------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| users              | id, name, email, password, points, avatar, bio                                               | HasMany: enrollments, courses, quiz_attempts, user_progress<br>BelongsToMany: badges (via user_badges) |
| courses            | id, teacher_id, title, description, category, level, is_published, duration_hours            | BelongsTo: users (teacher)<br>HasMany: lessons, enrollments, quizzes                                   |
| lessons            | id, course_id, title, description, video_url, content, duration_minutes, order, is_published | BelongsTo: courses<br>HasOne: quizzes<br>HasMany: user_progress                                        |
| enrollments        | id, user_id, course_id, progress_percent, completed_at                                       | BelongsTo: users, courses                                                                              |
| quizzes            | id, course_id, lesson_id, title, time_limit_minutes, passing_score, is_published             | BelongsTo: courses, lessons<br>HasMany: questions, quiz_attempts                                       |
| questions          | id, quiz_id, question_text, type, points, order                                              | BelongsTo: quizzes<br>HasMany: question_options, attempt_answers                                       |
| question_options   | id, question_id, option_text, is_correct, order                                              | BelongsTo: questions                                                                                   |
| quiz_attempts      | id, user_id, quiz_id, score, total_points, passed, started_at, completed_at                  | BelongsTo: users, quizzes<br>HasMany: attempt_answers                                                  |
| attempt_answers    | id, attempt_id, question_id, selected_option_id, is_correct                                  | BelongsTo: quiz_attempts, questions, question_options                                                  |
| user_progress      | id, user_id, lesson_id, is_completed, watch_percent, completed_at                            | BelongsTo: users, lessons                                                                              |
| badges             | id, name, description, icon, color, criteria_type, criteria_value                            | BelongsToMany: users (via user_badges)                                                                 |
| discussion_posts   | id, course_id, user_id, title, body, is_pinned                                               | BelongsTo: courses, users<br>HasMany: discussion_replies                                               |
| discussion_replies | id, post_id, user_id, body                                                                   | BelongsTo: discussion_posts, users                                                                     |

# Algorithm — Merge Sort Leaderboard

### Core Algorithm: Merge Sort (O(n log n))

The competitive leaderboard is ranked using a custom Merge Sort implementation in PHP, rather than relying on a simple database ORDER BY. This satisfies the algorithmic requirement and demonstrates efficient sorting with guaranteed O(n log n) time complexity regardless of input order.

### Implementation in StudentController.php

```
/** * Sort students by points descending using Merge Sort * Time Complexity: O(n log n) * Space Complexity: O(n) */private function mergeSort(array $students): array { if (count($students) <= 1) { return $students; } $mid = (int)(count($students) / 2); $left = array_slice($students, 0, $mid); $right = array_slice($students, $mid); $left = $this->mergeSort($left); $right = $this->mergeSort($right); return $this->merge($left, $right); } private function merge(array $left, array $right): array { $result = []; $i = $j = 0; while ($i < count($left) && $j < count($right)) { // Sort descending by points if ($left[$i]->points >= $right[$j]->points) { $result[] = $left[$i++]; } else { $result[] = $right[$j++]; } } while ($i < count($left)) { $result[] = $left[$i++]; } while ($j < count($right)) { $result[] = $right[$j++]; } return $result; }
```

### How It Is Used

```
public function leaderboard() { // Retrieve all students with their badge counts $students = User::role('student') ->withCount('badges') ->get()->toArray(); // Sort with custom Merge Sort (not SQL ORDER BY) $sorted = $this->mergeSort( User::role('student')->withCount('badges')->get()->all() ); return view('student.leaderboard', [ 'students' => $sorted, 'myRank' => collect($sorted)->search( fn($s) => $s->id === auth()->id() ) + 1, ]); }
```

### Complexity Analysis

| Metric       | Value      | Explanation                                                       |
|--------------|------------|-------------------------------------------------------------------|
| Best Case    | O(n log n) | Merge sort is not affected by input order — always divides evenly |
| Average Case | O(n log n) | Consistent performance across random inputs                       |
| Worst Case   | O(n log n) | Guaranteed even for reverse-sorted or identical input             |
| Space        | O(n)       | Auxiliary arrays created during merge step                        |
| Stability    | Stable     | Preserves original order for equal-points ties                    |

## Platform Features

### Student Portal (8 Views)

| Feature        | Route                            | Description                                                                              |
|----------------|----------------------------------|------------------------------------------------------------------------------------------|
| Dashboard      | /student/dashboard               | Enrolled courses, recent activity, points summary, badge highlights                      |
| Course Catalog | /student/courses                 | Browse all published courses with search, filter by category/level, and one-click enroll |
| Course Detail  | /student/courses/{id}            | Lesson list, progress bar, quiz status, discussion link, enrollment status               |
| Lesson Player  | /student/lessons/{id}            | Embedded YouTube/Vimeo player, lesson notes, Mark Complete button (+20 pts)              |
| Quiz           | /student/quizzes/{id}            | Timed MCQ quiz, question-by-question navigation with Alpine.js countdown timer           |
| Quiz Result    | /student/quiz-result/{attempt}   | Score, pass/fail, full answer review with correct answers, points earned                 |
| Leaderboard    | /student/leaderboard             | Merge sort-ranked table of all students by points, with personal rank highlight          |
| Badges         | /student/badges                  | Grid of all badges — earned (colored) and locked (greyed out) — with criteria            |
| Discussion     | /student/courses/{id}/discussion | Forum for each course, create posts (+5 pts), reply (+2 pts), view pinned posts          |

### Teacher Portal (11 Views)

| Feature           | Description                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------|
| Dashboard         | Stats on courses, total students, enrolments, quiz attempts, recent activity feed                     |
| Course Management | List, create, edit, delete courses. Publish/unpublish toggle.                                         |
| Lesson Management | Per-course lesson list, create/edit with video URL + rich content, reorder                            |
| Quiz Builder      | Create quizzes with time limit, passing score, and add multiple-choice questions with correct answers |
| Student Progress  | Per-course table of enrolled students with completion % and quiz scores                               |
| Discussions       | View and moderate all course discussions, pin important posts                                         |

### Admin Panel (7 Views)

| Feature           | Description                                                                           |
|-------------------|---------------------------------------------------------------------------------------|
| Dashboard         | Platform-wide stats: total users, courses, enrollments, revenue; activity log         |
| User Management   | Full CRUD on all users, role assignment (student/teacher/admin), account suspension   |
| Course Moderation | Publish/unpublish any course, view enrollment counts, delete                          |
| Reports           | Quiz pass rates, average scores, top students by points, course completion stats      |
| Badge Management  | Create badges with criteria type (points/lessons/quizzes), icon, color; delete badges |

# Gamification System

LearnHub's gamification engine is designed to motivate learners through visible rewards, friendly competition, and achievement recognition. Points accumulate across all learning activities and feed directly into the leaderboard ranking.

## Points Engine

| ACTION                   | POINTS AWARDED | TRIGGER                                                               |
|--------------------------|----------------|-----------------------------------------------------------------------|
| Complete a lesson        | +20 pts        | Student clicks "Mark Complete" on a lesson page                       |
| Pass a quiz              | +50 pts        | Quiz attempt submitted with score ≥ passing_score threshold           |
| Fail a quiz              | +10 pts        | Quiz attempt submitted below passing threshold (participation reward) |
| Create a discussion post | +5 pts         | Student submits a new forum post in a course discussion               |
| Reply to a post          | +2 pts         | Student submits a reply to an existing discussion post                |

## Badge System

Eight achievement badges are defined in the database, each with a criteria type and threshold value. Badges are awarded automatically when students meet the criteria through normal platform usage.

| BADGE           | CRITERIA TYPE     | THRESHOLD | DESCRIPTION                           |
|-----------------|-------------------|-----------|---------------------------------------|
| 🏆 First Steps   | lessons_completed | 1         | Complete your very first lesson       |
| 📖 Bookworm      | lessons_completed | 10        | Complete 10 lessons across any course |
| 🏆 Top Scorer    | points            | 500       | Accumulate 500 total points           |
| 🏆 Quiz Master   | quiz_perfect      | 1         | Score 100% on any quiz                |
| 🏆 Consistent    | enrollments       | 3         | Enrol in at least 3 different courses |
| 🏆 Contributor   | discussions       | 5         | Create 5 or more discussion posts     |
| 🎓 Graduate      | courses_completed | 1         | Fully complete an entire course       |
| 🏆 Elite Learner | points            | 1000      | Reach 1,000 total points              |

## Computed Model Attributes

```
// User model - dynamic avatar via UI Avatars API public function getAvatarUrlAttribute(): string { return $this->avatar ?? 'https://ui-avatars.com/api/?name=' . urlencode($this->name) . '&background=6366f1&color=fff&size=128'; } // Lesson model - convert YouTube/Vimeo URLs to embed URLs public function getEmbedUrlAttribute(): ?string { if (str_contains($this->video_url, 'youtube.com')) { return 'https://www.youtube.com/embed/' . $videoId; } if (str_contains($this->video_url, 'vimeo.com')) { return 'https://player.vimeo.com/video/' . $videoId; } return $this->video_url; } // QuizAttempt model - percentage score public function getScorePercentAttribute(): float { return $this->total_points > 0 ? round(($this->score / $this->total_points) * 100, 1) : 0; }
```

# Controllers & Routing

## Controller Architecture

| CONTROLLER        | MIDDLEWARE                              | RESPONSIBILITIES                                                                                                                                                                    |
|-------------------|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| StudentController | <div>auth</div> <div>role:student</div> | dashboard, courses, enroll, showCourse, watchLesson, markComplete, takeQuiz, submitQuiz, quizResult, leaderboard, discussion, createPost, createReply, badges                       |
| TeacherController | <div>auth</div> <div>role:teacher</div> | dashboard, courses (index/create/store/edit/update/destroy), lessons (create/store/edit/update/destroy), quiz (create/store/edit/update), addQuestion, studentProgress, discussions |
| AdminController   | <div>auth</div> <div>role:admin</div>   | dashboard, users (index/create/store/edit/update/destroy), courses (index/togglePublish/destroy), reports, badges (index/store/destroy)                                             |

## Route Structure (routes/web.php)

```
// Public routes Route::get('/', fn() => view('welcome')); // Shared dashboard redirect Route::get('/dashboard', fn() => redirect( auth()->user()->hasRole('admin') ? '/admin/dashboard' : (auth()->user()->hasRole('teacher') ? '/teacher/dashboard' : '/student/dashboard') ))->middleware('auth')->name('dashboard'); // Student routes (auth + role:student) Route::middleware(['auth', 'role:student'])->prefix('student') ->name('student.')->group(function() { ... }); // Teacher routes (auth + role:teacher) Route::middleware(['auth', 'role:teacher'])->prefix('teacher') ->name('teacher.')->group(function() { ... }); // Admin routes (auth + role:admin) Route::middleware(['auth', 'role:admin'])->prefix('admin') ->name('admin.')->group(function() { ... });
```

## Middleware Registration (bootstrap/app.php)

```
return Application::configure(basePath: DIR . '/../') ->withMiddleware(function(Middleware $middleware) { $middleware->trustProxies(headers: Request::HEADER_X_FORWARDED_FOR | ...); $middleware->alias([ 'role' => \Spatie\Permission\Middleware\RoleMiddleware::class, 'permission' => \Spatie\Permission\Middleware\PermissionMiddleware::class, ]); })->create();
```



# Deployment & Project Summary

## Startup Process (start.sh)



### Generate .env

Reads Replit secrets (PGHOST, PGPORT, PGDATABASE, PGUSER, PGPASSWORD) and writes a .env file with DB\_CONNECTION=pgsql and APP\_KEY.



### Clear Cache

Runs php artisan config:clear and cache:clear to ensure fresh configuration on every boot.



### Run Migrations

Executes php artisan migrate --force to create or update all 13 database tables.



### Storage Link

Creates the public/storage symlink (php artisan storage:link) for file uploads.



### Serve Application

Starts php artisan serve --host=0.0.0.0 --port=5000 to expose the app on Replit's public domain.

## Project Statistics

| Metric                  | Count                                                     |
|-------------------------|-----------------------------------------------------------|
| Database Migrations     | 13                                                        |
| Eloquent Models         | 13                                                        |
| Controllers             | 3 (StudentController, TeacherController, AdminController) |
| Blade Views             | 26 (8 student + 11 teacher + 7 admin + welcome + layout)  |
| Named Routes            | 50+                                                       |
| Seeded Users            | 13 (1 admin, 2 teachers, 9 students, 1 demo)              |
| Seeded Courses          | 4 (with 16 lessons, 2 quizzes, 9 questions)               |
| Gamification Badges     | 8                                                         |
| Lines of Code (approx.) | 5,000+                                                    |
| GitHub Repository       | github.com/anis3669/learnhub                              |

### Future Enhancements

Planned improvements include: email notifications for course completions and quiz results, file upload support for lesson materials (PDFs, slides), live Q&A sessions via WebSocket, certificate generation on course completion, Stripe payment integration for premium courses, and a mobile-responsive PWA version.